Technology Stack

Date	29 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5	Provides a responsive, user-friendly interface for farmers to enter soil and climate data and view crop recommendations.

2	**Backend / Server-Side**	Python, Flask	Flask provides a lightweight backend for handling user requests, integrating the machine learning model, and processing prediction logic efficiently.
3	**Database / Data Storage**	SQLite, CSV Dataset	SQLite stores user information and prediction history, while CSV datasets are used for machine learning model training and testing.
4	**Cloud / Hosting / Deployment**	Local Flask Server	Enables simple deployment, easy testing, and cost-effective hosting for the web application.
5	**Version Control & CI/CD**	GitHub	Maintains project versions, supports collaboration among team members, and tracks code changes efficiently.

6	**Third-Party APIs / Other Tools**	API, Scikit-learn, Pandas, NumPy, Matplotlib	API provides environmental data. Scikit-learn builds the prediction model. Pandas and NumPy process agricultural datasets. Matplotlib visualizes analysis results.